

How to Train Your Intercept

A Comparison of Intercept Update Strategies in Coordinate Descent

Johan Larsson ¹ Department of Mathematical Sciences, University of Copenhagen

Date published: 2025-07-20 Last modified: 2025-07-20

Abstract

Training intercepts in regularized regression is a common problem, yet it is often overlooked in the literature. This document explores different strategies for updating intercepts in coordinate descent algorithms, comparing gradient, Newton, and exact strategies.

Keywords: intercepts, coordinate descent, optimization, regularization

1 Contents

2	1 Introduction	1
3	References	1

4 1 Introduction

5 Intercept updating is a overlooked issue in the field of statistical learning. For regularized regression,
6 coordinate descent is the dominating algorithm, yet it is not clear how to best update the intercept
7 term. Implementations differ in how they handle this, for instance `skglm` uses a gradient step,
8 `LIBLINEAR` uses a Newton step, and `blitz` uses an exact update (based on Newton steps).

9 References

¹Corresponding author: jola@math.ku.dk

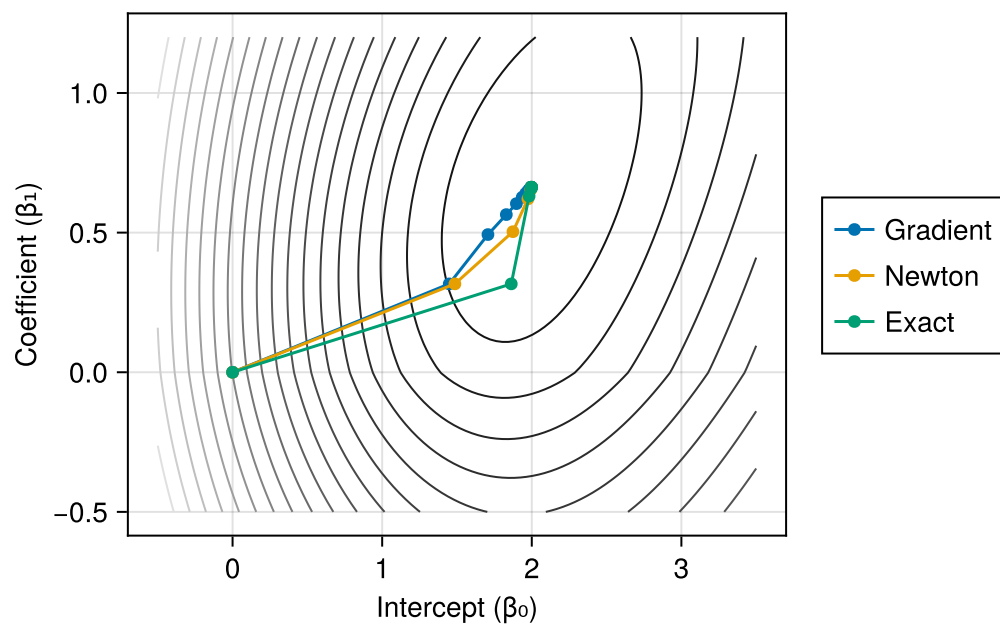


Figure 1: Parametric Plots